MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5644

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Voltage regulator, subminiature

Outline	---	3-4 (EIA)
Base	---	E8-10
Envelope	---	T3
Cathode	---	Glow discharge

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	a	k	int con	k	int con	int con	int con	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ez Total darkness	Ez Ambient light	Operating voltage	Operating current	TA	TE	Ⓔ Alt
Unit:	Vdc	Vdc	Vdc	mAdc	°C	°C	ft
Maximum:	---	---	110	25	---	+220	Note 6
Minimum:	175	130	80	5	-55	---	---
TEST CONDITIONS:	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 (see note 1)							
3347	Ionization voltage (1)	Ebb'Ib = 5 to 25 mAdc; illumination = 5 to 50 foot-candles (see note 2)	0.4	II	Ez	---	120	Vdc
3337	Voltage drop (1)	Ebb'Ib = 25 mAdc	0.4	II	Etd	85	105	Vdc
3337	Voltage drop (2)	Ebb'Ib = 5 mAdc	0.4	II	Etd	85	105	Vdc
(E) 3335	Regulation	Etd(2) minus Etd(1)	0.4	II	Reg	---	5.0	Vdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2 (see note 1)							
1031	Variable-frequency vibration	No voltages; post-shock and vibration-fatigue test end points apply	10.0	See note 3	---	---	---	---
3347	Ionization voltage (2)	Ebb'Ib = 5 to 25 mAdc (see note 4)	6.5	Code F	Ez	---	175	Vdc
3305	Leakage current	Eb = 50 Vdc	6.5	Code F	Lib	---	10	μAdc
3278	Oscillation test	Esig = 15 mVac; Ebb'Ib = 5 to 25 mAdc	2.5	I	---	---	---	---
3278	Noise test	Ebb'Ib = 25 mAdc	2.5	I	Eb	---	15	mvac
1116	Lead fatigue		2.5	Code G	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
1041	Shock	450 G	---	---	---	---	---	---
1031	Vibration-fatigue test	2.5 G; fixed frequency; F = 25 Hz (min), 60 Hz (max)	6.5	See note 3	---	---	---	---
---	Post-shock and vibration-fatigue test end points	Ionization voltage (1)	---	---	Ez	---	130	Vdc
		Regulation	---	---	Reg	---	±5.0	Vdc
		Voltage drop (1)	---	---	Etd	---	108	Vdc
		Voltage drop (2)	---	---	Etd	---	108	Vdc
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: 1ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: CUMULATIVE SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1521	Survival-rate life test	Intermittent life-test, or equivalent conditions; TA = room	---	---	---	---	---	---	---	---
---	Survival-rate life-test end point	Short and discontinuity detection	---	---	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	GROUP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1ST SAMPLE	CUMULATIVE SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
1501	Intermittent life test	Ebb/Eb = 25 mAdc; TE = +220°C (min) (see note 5)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Regulation	---	---	1	3	Reg	---	±5.0	Vdc
		Voltage drop (1)	---	---	1	3	Etd	82	108	Vdc
		Voltage drop (2)	---	---	1	3	Etd	82	108	Vdc
		Ionization voltage (1)	---	---	1	3	Ez	---	125	Vdc
		Total defectives	---	---	3	6	---	---	---	---
---	Intermittent life-test end points (1,000 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Regulation	---	---	1	3	Reg	---	±6.0	Vdc
		Voltage drop (1)	---	---	1	3	Etd	80	110	Vdc
		Voltage drop (2)	---	---	1	3	Etd	80	110	Vdc
		Ionization voltage (1)	---	---	1	3	Ez	---	130	Vdc
		Total defectives	---	---	4	8	---	---	---	---

NOTES:

(E) 1. Sampling procedure shall be in accordance with MIL-STD-105 with sample size determined by lot size, except the minimum sample size shall be as specified below. Use the AQL and inspection level specified for each individual test item to determine the minimum sample size code letter. The maximum lot tolerance percent defectives (LTPD), specified below, are for information purposes and are defined as the percent defective in the lot for which the probability of acceptance is 0.10 as obtained from the operating characteristic curves in MIL-STD-105.

AQL (percent defective)	Inspection level	Maximum LTPD	Minimum sample size code
0.4	II	2.7	L
2.5	I	12.9	H
6.5	S3	27.1	G

2. This test to be performed at the conclusion of the holding period.

(E) 3. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

4. Conditions for this test shall be those of ionization voltage (1), except testing shall be done in total darkness and the tube shall not have conducted or have been exposed to light for at least 24 hours prior to testing.

(E) 5. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-hand measurement (method 1226), will be satisfied if a tube having bogey Etd (±3 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

6. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Air Force - 17
DSA -ES

User activities:
Army - WC
Navy - AS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2484)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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